1 The classification of plants according to form and structure is

- **A** scientific.
- **B** morphology.
- **C** use.
- **D** life cycle.

2 The process by which mature differentiated cells reverse to meristematic activity to form callus is called

- **A** dedifferentiation.
- **B** differentiation.
- **C** cytodifferentiation.
- **D** redifferentiation.

3 The non-sister chromatids of homologous chromosomes twist around and exchange genetic segments with each other during

- **A** diplotene stage.
- **B** pachytene stage.
- **C** zygotene stage.
- **D** diakinesis stage.

4 The pairing of homologous chromosomes is called

- **A** tetrads.
- **B** crossing over.
- **C** synapsis.
- **D** termination.

Leptotine — condence
Zygotine — pairing up
Z Pachetyne — crossing over.
Diak — Separate.

5 **Figure 5** shows a cell.

P M A T C

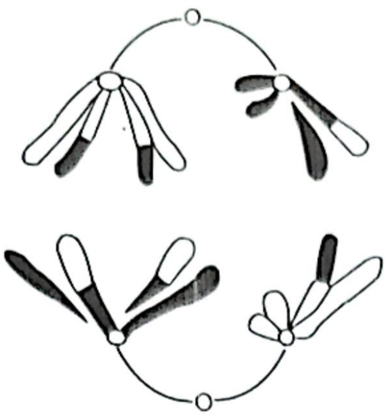

Fig. 5

Which stage of cell division is shown in **Figure 5**?

A Anaphase I of mitosis
B Anaphase I of meiosis
C Prophase I of mitosis
D Prophase I of meiosis

6 An angiosperm leaf carries **16** chromosomes. The number of chromosomes in its endosperm is

16

A 12.
B 16.
C 24.
D 32.

7 The role of exine in a pollen grain is to

A protect the pollen grain from the physical environment.
B produce the pollen tube nucleus.
C produce the generative nucleus.
D enhance the sticking of the pollen grain to the stigma.

8 Gibberellin controls

A flowering.
B germination in cereals.
C fruit formation without fertilisation.
D bud dormancy.

9 Plant meristematic cells have

A little cytoplasm.
B thick cell walls.
C large vacuoles.
D isodiametric shape.

10 Which is the major pathway for movement of water to the xylem vessels?

A symplast
B vacuolar
C cytoplasmic
D apoplast

11 The importance of hydrogen ions produced during photolysis of water is to

A reduce NADP to NADPH.
B reduce the charged photosystems.
C be taken up by electron carrier system.
D produce ATP from an inorganic phosphate and ADP molecule.

12 Farmers can reduce physiological dormancy in seeds by

A pricking the seed testa.
B soaking the seeds in a weak acid.
C soaking seeds in water for 24 hours.
D photoblasting.

13 The conversion of nitrates under anaerobic conditions into ammonia gas is

A mineralisation.
B volatilisation.
C nitrification.
D denitrification.

14 Which of the following colloids has the lowest cation exchange capacity?

A humus
B illite
C kaolinite
D montmorillonite

15 Which soil moisture is available for root absorption?

A field capacity.
B capillary
C hygroscopic
D gravitational

16 How can a farmer reduce bulk density of soil?

A practising monoculture
B use of heavy machinery
C adding anthill to the soil
D maintaining the same planting depth

17 Why are legumes commonly used as green manure? They have a

A moderate C/N ratio.
B high C/N ratio.
C low C/N ratio.
D fluctuating C/N ratio.

18 Potassium is commonly deficient in crops because it is

A easily leached.
B immobilised by organisms.
C fixed between layers of clay minerals.
D strongly adsorbed on clay collois.

19 An organic base + 5 C sugar produces

A DNA.
B RNA.
C nucleoside.
D nucleotide.

20 Selection of homozygous plants is

A mass selection.
B pureline selection.
C hybridisation.
D introduction.

21 A hormone pair required for a callus to differentiate are

A auxin and cytokinin.
B auxin and ethylene.
C auxin and absciccic acid.
D cytokinin and gibberellin.

22 Which of the following is a physical non-ionizing mutagen?

A ethylenimine
B ultraviolet light
C diethylsulphate
D gamma rays

23 How many sets of chromosomes are inherited from the maternal parent plant?

A $\frac{1}{2}$

B $\frac{1}{4}$

C 1

D 2

24 Which of the following mechanisms contribute to genetic variation in plants?

A independent assortment; crossing over; random fertilisation
B crossing over; cross pollination, self pollination
C cross pollination; hybridization; selfing
D pureline breeding; hybridization; crossing over

25 Which of the following is a parasitic weed in cereal crops?

A Pigweed
B Witch weed
C Black jack
D Upright starbur

26 Destroying volunteer plants in pest control helps to

A prevent the spread of pests to uninfected areas.
B destroy an alternative habitat for the pest when the crop is out of season.
C increase opportunity of a crop to grow large before pest infestation.
D increase the chances of spread of the same pest to the next crop.

27 Calculate the amount of chemical pesticide to add to a 16 *l* knapsack sprayer if the application rate is **500 ml/ha** and a plot measuring **50 × 100 m** needs application.

A 100 ml
B 250 ml
C 500 ml
D 1 000 ml

500ml = 1ha 10000
5000ml = 5000

28 Allelopathic plants control weeds by

A depriving weeds of light.
B competing for nutrients.
C competing for water.
D releasing chemicals which kill weeds.

29 The rationale of strip harvesting in controlling pests in the field is to

A disrupt survival of the pest in its habitat.
B remove pest infected plant parts.
C prevent mass migration of pests to neighbouring fields.
D promote healthy crops which are less susceptible to pest attack.

30 Primary tillage involves

A finer operations performed on the soil.
B crushing the clods and uprooting the remaining weeds.
C turning the soil and uprooting deep rooted plants.
D smoothening the soil surface and compacting the soil.

31 Identify the type of farming done in Natural Farming region III.

A Intensive farming
B Semi-extensive farming
C Diversified intensive farming
D Semi-intensive mixed farming

32 Which conditions predispose stored produce to damage by pests and diseases?

A High temperature and low moisture content.
B High temperature and high moisture content.
C Low temperature and low moisture content.
D Low temperature and high moisture content.

33 Select one crop rotation principle that reduces soil compaction.

A Alternate deep-rooted with shallow-rooted crops.
B Alternate crops with high root biomass and those with low root biomass.
C Include green manures and catch crops.
D Alternate nitrogen-fixing crops with high nitrogen demanding crops.

34 Which of the following is the effect of high plant population?

A Increase in cob size.
B Increase in weed density.
C Reduction in soil erosion.
D Reduction in plant height.

35 Which of the following strategies would a farmer adopt to conserve soil moisture?

A installing an irrigation system.
B adding organic matter to the soil.
C filling the soil more thoroughly and frequently.
D increasing the amount of chemical fertilisers used.

36 Identify an effective technique used to reduce soil erosion in an arable land.

A deep burial of stover
B cultivating less than 30 m from a stream
C bare waterway
D planting grass on contours

37 Identify an appropriate definition of multiple cropping.

A When a field is used to grow two or more crops.
B When a field has each row alternating with different crops.
C When a field is used to grow two or more crops throughout the year so that the field constantly has something growing on it.
D When a field is used to grow two or more crops during the growing season so that the field has something growing on it during the whole season.

38 The amount of residue left after harvesting mainly depends on

A time of harvesting the crop.
B method of harvesting the crop.
C time taken to reach maturity.
D yield obtained at harvesting.

39 Conservation agriculture on slopes can be used in association with

A contour grass strip.
B closer plant spacing.
C cover crops.
D organic mulch .

40 Soil temperature is influenced by

A residue.
B fertility.
C drainage.
D porosity.